「Exercise.

20.E Define a map

$$f:\mathbb{R}\to\mathbb{R} : x\mapsto \begin{cases} x & \text{if } x\notin\mathbb{Q} \\ 1-x & \text{if } x\in\mathbb{Q} \end{cases}$$

Then f is continuous only at $x=\frac{1}{2}$.

Sol). Let $\varepsilon>0$ be given. Taking $\delta=\varepsilon$. Then for any $x\in(\frac{1}{2}-\delta,\frac{1}{2}+\delta)$,

$$\left|f(x)-\frac{1}{2}\right| = \begin{cases} \left|x-\frac{1}{2}\right| & \text{if } x\notin\mathbb{Q} \\ \left|1-x-\frac{1}{2}\right| & \text{if } x\in\mathbb{Q} \end{cases}$$

Both case satisfies $\left|f(x)-\frac{1}{2}\right|<\varepsilon$. Therefore continuous.

Let $x\in\mathbb{R}$ with $x\neq\frac{1}{2}$. If $x\in\mathbb{Q}$, then by density of rationals, we can construct $\{x_n\}$ in $\mathbb{R}\setminus\mathbb{Q}$ s.t. $x_n\to x$ as $n\to\infty$

But $f(x_n)=x_n\to x$ as $n\to\infty$, not converges to $f(x)=1-x$.

Similarly, if $x\notin\mathbb{Q}$, it can be shown by the same process.

20.F If a continuous map $f:\mathbb{R}\to\mathbb{R}$ satisfies

$$f(x)=0 \quad \text{for all } x\in\mathbb{Q}$$

Then f is zero map.

Proof. Suppose that f is not zero map. Given $x\in\mathbb{R}$ satisfying $f(x)\neq 0$, take $\varepsilon=|f(x)|$.

For any $\delta>0$, by the density of rationals,

$$\exists t\in(x-\delta,x+\delta) \text{ s.t. } |f(x)-f(t)|=|f(x)|>\frac{1}{2}|f(x)|.$$

It is contradiction.

20.G Let $f,g:\mathbb{R}\to\mathbb{R}$ be continuous map. If $f(x)=g(x)$ $\forall x\in\mathbb{Q}$, then $f=g$.

Proof. Take $h=f-g$ and apply 20.F.

20.H. Using $|\sin x| \le |x|$, show sine function is continuous at $x=0$.
Moreover, using

$$\sin x - \sin u = 2\sin\tfrac{1}{2}(x-u)\cos\tfrac{1}{2}(x+u)$$

Show sine function continuous on $\mathbb{R}$.

proof. Given $\varepsilon>0$, take $\delta=\varepsilon$. Then,

$$|x|<\delta \Rightarrow |\sin x| \le |x| < \delta = \varepsilon$$

Thus, continuous at $x=0$. Similarly, taking $\delta=\varepsilon$ then

$$|x-u|<\delta \Rightarrow |\sin x - \sin u| = \left|2\sin\tfrac{1}{2}(x-u)\cos\tfrac{1}{2}(x+u)\right| \le |x-u| < \varepsilon.$$

20.J. Define a map

$$g:\mathbb{R}\to\mathbb{R}: x \mapsto \begin{cases} x\cdot\sin\left(\frac{1}{x}\right) & \text{if } x\neq 0 \\ 0 & \text{if } x=0 \end{cases}$$

Then g is continuous on $\mathbb{R}$.

proof. Since x, $\sin x$, $\frac{1}{x}$ are continuous if $x\neq 0$, so is $x\sin\frac{1}{x}$.
Given $\varepsilon>0$, put $\delta=\varepsilon$. Then

$$0<|x|<\delta \Rightarrow \left|x\sin\left(\tfrac{1}{x}\right)\right| \le |x| < \varepsilon$$

Therefore, g is continuous everywhere.

20.K. Define a map

$$F:\mathbb{R}^2\to\mathbb{R}: (x,y)\mapsto \begin{cases} x^2+y^2 & \text{if } (x,y)\in\mathbb{Q}\times\mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$$

Determine the points where F is continuous.

Sol) Consider $(0,0)$. Given $\varepsilon>0$, taking $\delta=\sqrt{\varepsilon}$

$$(x,y)\in B_\delta(0,0) \Rightarrow |f(x,y)| \le x^2+y^2 < \delta^2 = \varepsilon.$$

Thus conti. at zero. But, if $(x,y)\neq 0$,